

Derivation and Validation of a Prognostic Model for Pulmonary Embolism

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Abstract

Rationale

An objective and simple prognostic model for patients with pulmonary embolism could be helpful in guiding initial intensity of treatment.

Objectives

To develop a clinical prediction rule that accurately classifies patients with pulmonary embolism into categories of increasing risk of mortality and other adverse medical outcomes.

Methods

We randomly allocated 15,531 inpatient discharges with pulmonary embolism from 186 Pennsylvania hospitals to derivation (67%) and internal validation (33%) samples. We derived our prediction rule using logistic regression with 30-day mortality as the primary outcome, and patient demographic and clinical data routinely available at presentation as potential predictor variables. We externally validated the rule in 221 inpatients with pulmonary embolism from Switzerland and France.

Measurements

We compared mortality and nonfatal adverse medical outcomes across the derivation and two validation samples.

Main Results

The prediction rule is based on 11 simple patient characteristics that were independently associated with mortality and stratifies patients with pulmonary embolism into five severity classes, with 30-day mortality rates of 0–1.6% in class I, 1.7–3.5% in class II, 3.2–7.1% in class III, 4.0–11.4% in class IV, and 10.0–24.5% in class V across the derivation and validation samples. Inpatient death and nonfatal complications were $\leq 1.1\%$ among patients in class I and $\leq 1.9\%$ among patients in class II.

Conclusions

Our rule accurately classifies patients with pulmonary embolism into classes of increasing risk of mortality and other adverse medical outcomes. Further validation of the rule is important before its implementation as a decision aid to guide the initial management of patients with pulmonary embolism.

[mortality](#), [prognosis](#), [pulmonary embolism](#)

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